

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

PRODUCT DESIGN & DEVELOPMENT

(Common to CE, EEE, ECE, CSE, IT, FT, AI&DS, CSE (AI), CSE (DS), CSE (AI&ML), CSE (IOT), CSE (CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) What are the steps involved in creating a network plan? | 2M |
| (b) Write a short note on Simultaneous Engineering. | 2M |
| (c) What is a Requirement List? | 2M |
| (d) Differentiate between anonymous customers and specific customers. | 2M |
| (e) What is the importance of requirements list in conceptual design? | 2M |
| (f) Explain the necessity of breaking a function into sub functions. | 2M |
| (g) Define Ergonomics. | 2M |
| (h) Give one example each for Self-help and self-damaging principles. | 2M |
| (i) Write a short note on Mechanical Connections. | 2M |
| (j) What characteristic difference exists between concepts of Mechatronics and Adaptronics? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 With the help of neat sketch discuss about the steps in Product Design and Development Process. 10M
- OR**
- 3 (a) What elements determine the success or failure of the product in the market? Discuss. 5M
(b) List and Explain the skills required for Project Manager? 5M
- 4 (a) What is a Check List? How it plays an important role in preparing the requirements list. 5M
(b) Depict the format of the requirements list with a practical application. 5M
- OR**
- 5 (a) How can the product requirements change during the product creation and product use? 5M
(b) Discuss how the product requirements list compiled from Partial Requirements list in product design and development process? 5M
- 6 With the help of neat sketch, explain about various steps in the conceptual design? 10M
- OR**
- 7 (a) What are the steps followed for establishing function structures? 5M
(b) During the Conceptual phase, the evaluation of technical and economic characteristics should be considered as early as possible. Why? 5M
- 8 What is Design for Ergonomics? What characteristics are used to identify Ergonomics requirements? 10M
- OR**
- 9 What are the basic rules that are to be followed in embodiment design? Explain them in detail. 10M
- 10 (a) Discuss about the working principle and structural characteristics of Material Connections. 5M
(b) Give the applications of Elastic force connections. 5M
- OR**
- 11 (a) Discuss about the basic structure of a Adaptronics. 5M
(b) What are the goals and Limitations of Adaptronics? 5M
